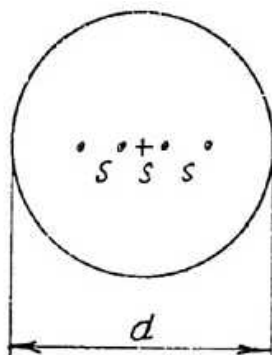


I. THIN, CIRCULAR SLICE.I.1) Measurement in the Center.thickness $t < \frac{s}{2}$

This case has been treated by Smits (e). The result is:

$$\rho = G \frac{V}{I}, \quad G = \frac{\pi}{\ln 2} \cdot t \cdot C_0\left(\frac{d}{s}\right) = 4,5324 \cdot t \cdot C_0\left(\frac{d}{s}\right) \quad (18)$$

where:

$\frac{\pi}{\ln 2} \cdot t = 4,5324 \cdot t$ is the geometric factor for an infinitely large, thin slice (section D.2)

$$C_0\left(\frac{d}{s}\right) = \frac{1}{1 + \frac{1}{\ln 2} \ln \left[\frac{1 + 3\left(\frac{s}{d}\right)^2}{1 - 3\left(\frac{s}{d}\right)^2} \right]} \quad (19)$$

is the additional

correction for a finite diameter $C_0 \rightarrow 1$ as $d/s \rightarrow \infty$.

The magnitude $\frac{\pi}{\ln 2} \cdot C_0 = 4,5324 \cdot C_0$ has been tabulated by Smits (e) and later more extensively by Swartzendruber (h). $C_0\left(\frac{d}{s}\right)$ is shown at page 39 and $\frac{\pi}{\ln 2} \cdot C_0\left(\frac{d}{s}\right)$ at page 40.